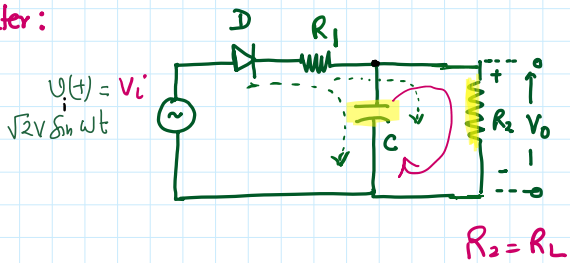
Half Wave Rectification with Filter:



$V_C = V_0$, $R_1 \ll R_2$

$R_2 = R_L$

D conducts when $V_i > V_C$

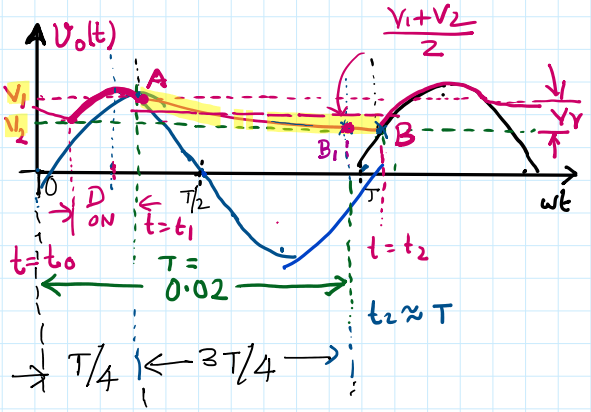
D does not conduct when $V_i < V_C$

$\tau = R_2 C > T$ (T: Time period of V_i)

$R_L = R_2 = 10\ k\Omega$
$C = 10\ \mu F$

$\tau = R_2 C = 10 \times 10 \times 10^{-3} = 0.1$ Sec

$\tau > T$ [for 50 Hz input signal, $T = \frac{1}{50} = 0.02 = 20$ msec.]



$V_2 = V_1 e^{-t/R_2C}$

$V_1 \simeq V_m$ (for all practical purposes)

$V_2 \simeq V_m e^{-t/R_2C} = V_m e^{-t/0.1} = V_m e^{-10t}$

We can take initial time as $t = t_0 = T/4$

Then, $V_2 = V_m e^{-(t-T/4)/R_2C}$

Point B can be approximated by B' as V_2 magnitude does not change much.

Thus considering $t = T$,

$V_2 = V_m e^{-(0.02 - 0.005)/0.1} \simeq 0.86\, V_m$

The ripple in output voltage is $V_r = \Delta V_0 = V_m - 0.86\, V_m$

$= 0.14 V_m$

$= 14\%$ of the peck value

There we define, two important terms: 1. Ripple factor (RF)
2. Form factor (FF)

The ripple factor (RF) is defined as,

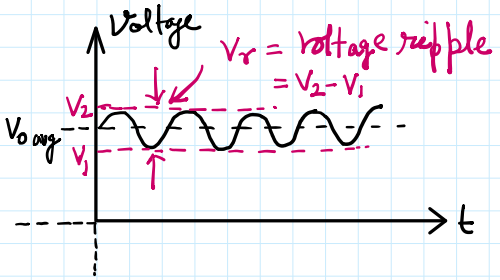
$$RF = \frac{\text{Rms value of the voltage ripple}}{\text{Average value of the voltage}}$$

The form factor (FF) is defined as

The form factor (FF) is defined as

$$FF = \frac{\text{Rms value of the voltage}}{\text{Average value of the voltage}}$$

And there is relation between them, as described below.



$$v_o \simeq V_{oavg} + V_{r\,max} \sin \omega t$$

$$V_{orms} = \left[\frac{1}{T}\int_0^T v_o^2(t)\,dt\right]^{1/2} = \left[\frac{1}{T}\int_0^T (V_{oavg} + V_{r\,max}\sin\omega t)^2\,dt\right]^{1/2}$$

$$\downarrow$$

$$V_{orms}^2 = \left[\frac{1}{T}\int_0^T V_{oavg}^2\,dt + \frac{1}{T}\int_0^T (V_{r\,max}\sin\omega t)^2\,dt\right]$$

$$= V_{oavg}^2 + \left(\frac{V_{r\,max}}{\sqrt{2}}\right)^2 = V_{oavg}^2 + V_{r\,rms}^2$$

Thus $V_{orms} = \sqrt{V_{oavg}^2 + V_{r\,rms}^2}$

Divide both sides by V_{oavg}, we get

$$\frac{V_{orms}^2}{V_{oavg}} = \frac{V_{oavg}^2}{V_{oavg}^2} + \frac{V_{r\,rms}^2}{V_{oavg}^2}$$

$$(FF)^2 = 1 + (RF)^2$$

$$RF = \sqrt{(FF)^2 - 1}$$

Just compare the Ripple factor for two cases:

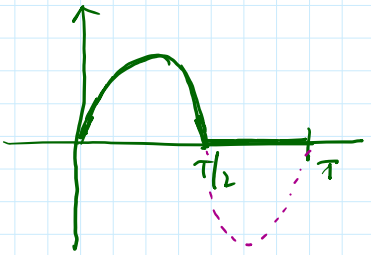
(1) without C-filter

(2) with C-filter

1. Without C-filter :

$V_{o,avg} = \frac{V_m}{\pi}$

$V_{o,rms} = \frac{V_m}{2}$



Average Value

$$V_{o,avg} = \frac{1}{2\pi}\int_0^{2\pi} v_o(\omega t)\, d(\omega t)$$

$$= \frac{1}{2\pi}\left[\int_0^{\pi} V_m \sin\omega t\, d(\omega t) + \int_{\pi}^{2\pi} 0\, d(\omega t)\right]$$

$$\boxed{V_{o,avg} = \frac{V_m}{\pi}}$$

RMS Value

$$V_{o\,rms} = \left[\frac{1}{2\pi}\int_0^{2\pi}(V_m \sin\omega t)^2\, d(\omega t)\right]^{1/2}$$

$$= \left\{\frac{V_m^2}{2\pi}\left[\int_0^{\pi}\left(\frac{1-\cos 2\omega t}{2}\right) d(\omega t) + \int_{\pi}^{2\pi} 0\, d(\omega t)\right]\right\}^{1/2}$$

$$= \left[\frac{V_m^2}{2\pi}\int_0^{\pi}\left\{\frac{(1-\cos 2\omega t)}{2}\right\} d(\omega t)\right]^{1/2}$$

$$V_{o\,rms} = \frac{V_m}{2}.$$

Thus the form factor and ripple factor are given as following.

$$FF = \frac{V_{orms}}{V_{oavg}} = \frac{V_m/2}{V_m/\pi} = \pi/2 = 1.57$$

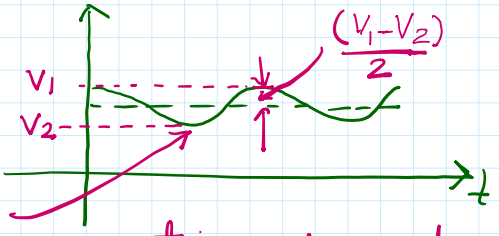
$$RF = \sqrt{FF^2 - 1} = \sqrt{1.57^2 - 1} = 1.21$$

2. With Filter Capacitor :

As we have computed $V_r = (V_1 - V_2) = V_m - 0.86V_m = 0.14V_m$

$$V_{r,rms} = \frac{(V_1 - V_2)}{2}\cdot\frac{1}{\sqrt{2}}$$

$$= \frac{0.14V_m}{2\sqrt{2}} = 0.05\,V_m$$



approximating ripple variation as sinusoidal variation.

$$V_{o,avg} \simeq \frac{V_1 + V_2}{2} = \frac{V_m + 0.86V_m}{2}$$

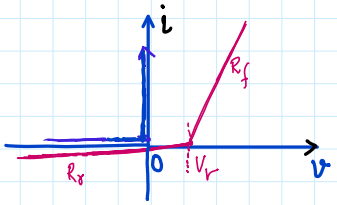
$$= 0.93V_m$$

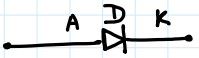
$$RF = \frac{V_{r,rms}}{V_{oavg}} = \frac{0.05V_m}{0.93V_m} = 0.053$$

Earlier without Capacitor

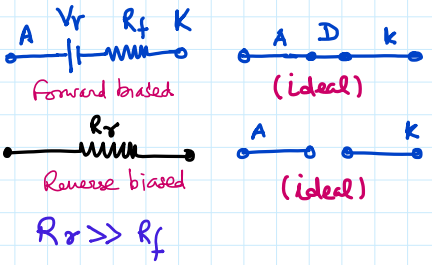
$$RF = \frac{V_{r,rms}}{V_{o\,avg}} = \frac{0.05\,V_m}{0.93\,V_m} = 0.053$$

Earlier without Capacitor
RF = 1.21
Reduction by $\frac{1}{23}$ th.





≡



$R_r >> R_f$

Piece-wise linear VI characteristic of an unction diode
